TECHNOLOGY AND ENVIRONMENT (TE)

MSE 2: E-Waste Management & Circular Economy Case Studies

Case Study 1: Why Japan Recycles More Than Most

BACKGROUND & STRATEGY

- **Shared responsibility:** Consumers, retailers, manufacturers.
- **Recycling Laws:** Large appliances (2001) & Small devices (2013).
- **Consumer fee:** Paid explicitly at the point of disposal.

IMPACT & RESULTS

- Overall recycling rate of **19.5% (FY 2023)**.
- **3.63 million** smartphones collected safely.
- FY 2030 targets: **70% lithium**, **95% nickel/cobalt**.

Strong legal framework.

Low small-device collection.

Critical mineral recovery.

Lithium battery fires.



Case Study 2: Switzerland's Success Story



BACKGROUND & STRATEGY

- **ORDEE 1998:** Mandatory national take-back system.
- **Advance Recycling Fee (ARF):** Charged at time of purchase.
- **PRO model:** SENS eRecycling & Swico operations.

IMPACT & RESULTS

- Total recycled: **~144,000 tonnes (2025)**.
- Per capita recycling: **11.48 kg/resident**.
- Valuable materials recovered: **70,000 tonnes**.

Strong PRO funding.

High operational costs.

Circular design potential.

Commodity volatility.

INDIA TAKEAWAY

Pilot advance recycling fee + strengthen retailer take-back.

Case Study 3: India’s E-Waste Management Rules 2022

BACKGROUND & STRATEGY

- Effective: **1 April 2023**.
- **Centralized EPR portal** administered by CPCB.
- **EPR certificates** purchased from registered recyclers.
- Local bodies responsible for orphan e-waste.

IMPACT & RESULTS

- Bengaluru registered recyclers: **49**.
- India e-waste generation: **1.7 million tonnes/year**.
- Growth rate: **15–20% annually**.
- Fake EPR certificates detected: **0.7 million**.

SWOT ANALYSIS

Comprehensive legal mandate.

Weak enforcement mechanisms.

Eco-Park development.

Informal sector dominance.

INDIA TAKEAWAY

Strengthen MRV + third-party audits + product-specific targets.

Case Study 4: India’s EPR – Success or Paper?

BACKGROUND & STRATEGY

- **Digital Tracking:** Producers register on CPCB portal.
- **Trading System:** Purchase EPR certificates from recyclers.
- **Compliance:** Quarterly / annual returns filed.

IMPACT & RESULTS

- Informal sector share: **80–90%**.
- Formal collection in Bengaluru: **~15%**.
- Fake vs. actual capacity gap: **38 times**.

INDIA TAKEAWAY

Move from paper compliance to funded, traceable EPR.

Digital compliance tracking.

Certificate verification fraud.

Blockchain tracking integration.

High consumer apathy.



Case Study 5: Bengaluru – India's E-Waste Capital?



BACKGROUND & STRATEGY

- **IT Hub:** Exceptionally high gadget penetration.
- **Formal network:** 33 registered recyclers.
- **Logistics:** GPS/RFID equipped collection vehicles.

IMPACT & RESULTS

- Annual generation: **~12,000 tonnes**.
- Recycling capacity: **75,020.92 MT/annum**.
- Formal collection rate: **~15%**.

High recycler concentration.

Low collection rates.

Corporate partnerships.

Rural e-waste influx.

INDIA TAKEAWAY

Add ward-level drop boxes + repair-shop incentives.

Case Study 6: Mumbai’s Ecosystem

BACKGROUND & STRATEGY

- **Scale:** India’s largest e-waste generator.
- **Policy:** BMC E-Waste Management Policy approved.
- **Infrastructure:** Major authorized collection centres.

IMPACT & RESULTS

- Annual generation: **120,000 MT**.
- Ecoreco Li-ion capacity: **6,000 MTPA**.
- Collected in 5 months: **21,570 kg**.

Dedicated municipal policy.

Fragmented collection network.

Li-ion recycling capacity.

Redevelopment disruptions.



INDIA TAKEAWAY

Fast-track BMC policy + expand ward-level collection.

Case Study 7: Kerala’s Clean Kerala Company



BACKGROUND & STRATEGY

- **Agency model:** Clean Kerala Company Ltd (CKCL).
- **Volunteers:** Haritha Karma Sena door-to-door collection.
- **Infrastructure:** Eco Banks as collection centres.

IMPACT & RESULTS

- Non-hazardous collected: **2,678 MT (2014–25)**.
- Hazardous e-waste: **161 MT**.
- 2025 ULB drive generated: **97,678 kg**.

Dedicated state agency.

Low public acceptance.

Scrap dealer integration.

Inter-state illegal dumping.

INDIA TAKEAWAY

Revitalize Eco Banks + localized incentive schemes.

Case Study 8: Pune's SWaCH Model

BACKGROUND & STRATEGY

- **Cooperative:** India's first waste picker-owned network.
- **Service:** Door-to-door collection with user fees.
- **Partnership:** Formal agreement with Pune Municipal Corp.

IMPACT & RESULTS

- Recyclables diverted: **8.3 crore kg/year**.
- Waste pickers empowered: **3,652 (70% women)**.
- PMC savings: **₹25+ crore/annum**.

INDIA TAKEAWAY

Replicate SWaCH model + expand into e-waste collection.

Worker cooperative ownership.

Social protection gaps.

E-pushcart pilot testing.

Climate-related disruptions.



Case Study 9: Dharavi Recycling Ecosystem



BACKGROUND & STRATEGY

- **Informal Hub:** Processes 600+ tons/day.
- **Value Chain:** Pickers → traders → micro-enterprises.
- **Sorting:** Specialized sorting of 12–20 types of plastic.

IMPACT & RESULTS

- Mumbai's plastic processed: **60–80%**.
- Waste pickers supported: **Up to 250,000**.
- Annual economic activity: **$1+ billion**.

Massive employment generation.

Severe health hazards.

EPR value-chain integration.

Displacement via redevelopment.

INDIA TAKEAWAY

Formalize through cooperatives + integrate into EPR chains.

Case Study 10: Dell's Circular Economy Strategy

BACKGROUND & STRATEGY

- **Closed-loop:** Advanced recycling programs.
- **Global network:** Comprehensive take-back systems.
- **2030 Moonshot:** 1:1 reuse/recycle ratio.

IMPACT & RESULTS

- Equipment recovered: **1.1 billion kg**.
- Sustainable packaging: **94.5% (FY2023)**.
- Recycled content: **14% (FY2024)**.

INDIA TAKEAWAY

Expand take-back + integrate informal sector + pilot closed-loop.

Global take-back network.

Recycled content gaps.

EPR partnerships.

Regulatory fragmentation.



Overall Takeaways for India

Strategic implementations required for a sustainable circular economy

Advance Fee

Implement visible fees at purchase for dedicated funding.

Specific Targets

Move beyond generic category targets to precise recovery metrics.

Formalization

Organize informal workers through funded cooperatives.

Digital MRV

Deploy blockchain tracking to halt EPR certificate fraud.

Ward Collection

Build localized drop-box infrastructure in urban centres.

Mineral Recovery

Prioritize extracting high-value materials from electronics.

Awareness

Execute public campaigns for households and institutions.

Eco-Parks

Link informal workers to formal recycling ecosystems.

Image Sources



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